



University of Milan
Data Science for Economics

Maven Rewards Challenge

Statistical Learning Project

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Contents

1	Introduction	2
2	Data Description	2
3	Data Manipulation	2
3.1	Data Cleaning	3
3.2	Feature Engineering	3
4	Unsupervised Part	4
4.1	Data Preparation	4
4.2	PCA and Hierarchical Clustering (for model selection)	6
4.3	K-Means	9
5	Supervised Learning	11
5.1	Data Preparation	11
5.2	Model Benchmarking and Selection	12
5.3	Final Model Evaluation	13
6	Conclusion and Findings	13

1 Introduction

The goal of this project is to understand customer behavior and extract insights that can support business decisions. In a competitive market, knowing how different customers act and what drives their loyalty is key to improving marketing performance and profitability. This study uses data analysis to identify customer patterns, predict their actions, and suggest strategies to strengthen engagement and retention.

The work started with data preprocessing and exploration. Missing and inconsistent values were corrected, and new features were created to capture useful behavioral details such as spending and visit frequency. After preparing the data, unsupervised learning methods were applied to find groups of customers with similar behaviors. These clusters were examined to understand their size, spending level, and overall importance for the business.

In the next step, supervised learning models were built to predict customer responses and highlight the main factors that influence engagement. Combining both unsupervised and supervised approaches gave a broader picture: who the customers are, how they behave, and how likely they are to respond in the future. The results of this project can guide better marketing decisions, helping the company target the right segments and design more effective retention programs.

2 Data Description

This project uses three relational, flat CSV datasets from Maven Analytics: *customers*, *offers*, and *events* together with an official *data_dictionary*. The *customers* file holds one record per customer with relatively stable attributes (e.g., demographics or registration metadata). The *offers* file lists marketed items and their descriptors (such as category, channel, value, and validity windows). The *events* log contains time-stamped customer interactions (e.g., redemptions and purchases). These files are designed to integrate through identifier fields where available (for example, *customer_id* in *events* aligning to *customers*, and *offer_id* aligning to *offers*), enabling both cross-sectional summaries and longitudinal analyses.

The *events* log provides the temporal backbone of the analysis: timestamps allow sequencing interactions, building recency/frequency measures, and deriving lead/lag features around offer exposure and redemption. Meanwhile, the *customers* table supplies the entity-level context required for grouping, cohorting, and stratification, and *offers* captures the characteristics of the offer portfolio that may explain heterogeneous engagement or conversion outcomes. Together, these tables support standard analytics workflows (descriptive profiling, cohort tracking, uplift/propensity modeling) once joined on shared identifiers.

For transparency and reproducibility, we reproduce the vendor-provided field definitions as a single consolidated data dictionary shown in table 1. This table reports, for each dataset, every column name, its declared type (as provided) and the textual description. We rely on these canonical definitions to guide feature engineering (e.g., selecting categorical vs. continuous fields), to check join keys and expected cardinalities, and to document any deviations introduced during preprocessing (such as type coercions or derived variables).

3 Data Manipulation

This section details how the raw *customers*, *offers*, and *events* files were transformed into analysis-ready tables. We first clean and integrate the sources, parsing semi-structured fields and enforcing consistent types and keys. We then engineer customer-level behavioral features (counts, rates, timings, and spend metrics), and assemble final datasets for unsupervised exploration and supervised learning.

Table 1: Data dictionary (verbatim from Maven Analytics release)

Table	Field	Description
offers	Table Description	Details on the offers sent to customers during the 30-day period
offers	offer_id	Unique offer ID (primary key)
offers	offer_type	Type of offer: bogo (buy one, get one), discount, or informational
offers	difficulty	Minimum amount required to spend in order to be able to complete the offer
offers	reward	Reward (in dollars) obtained by completing the offer
offers	duration	Days a customer has to complete the offer once they have received it
offers	channels	List of marketing channels used to send the offer to customers
customers	Table Description	Demographic data for each member
customers	customer_id	Unique customer ID (primary key)
customers	became_member_on	Date when the customer created their account (yyyymmdd)
customers	gender	Customer's gender: (M)ale, (F)emale, or (O)ther
customers	age	Customer's age
customers	income	Customer's estimated annual income, in USD
events	Table Description	Data on customer activity, with records for transactions, offers received, offers viewed, and offers completed
events	customer_id	Customer the event is associated with (foreign key)
events	event	Description of the event (transaction, offer received, offer viewed, or offer completed)
events	value	Dictionary of values associated with the event (amount for transactions, offer_id for offers received and viewed, and offer_id & reward for offers completed)
events	time	Hours passed in the 30-day period (starting at 0)

3.1 Data Cleaning

All processing is implemented in R using `dplyr/tidyr`. We begin by inspecting schema and row samples and then parsing the semi-structured `value` column in *events*. Specifically, we extract (i) `offer_id` for offer-related events, robust to both spellings '`offer_id`' and '`offer id`'; (ii) `reward` exclusively for `offer completed` events; and (iii) `amount` exclusively for `transaction` events. Membership dates in *customers* (`became_member_on`) are converted from `YYYYMMDD` to `Date`. Next, we integrate the sources by keys: *events* is left-joined to *customers* on `customer_id`, and the result is left-joined to *offers* on `offer_id`. The combined table is ordered by `customer_id` and event `time`. To preserve attribution, we propagate a missing `offer_id` forward to a `transaction` only when (a) the previous event for that customer is `offer completed` and (b) that previous row has a valid `offer_id`; this conservative rule avoids linking unrelated purchases to offers. Finally, we compute intermediate customer-level summaries and handle missingness: behavior-derived fields for inactive users are set to zero at the end of feature creation, while rows with missing core demographics (e.g., `income`) are removed where a complete-case cohort is required for downstream modeling.

3.2 Feature Engineering

From the integrated event log, we construct customer-level features. Engagement counts include `num_offers_received`, `num_offers_viewed`, and `num_offers_completed`. We derive rates

Table 2: Engineered variables (customer level) and construction rules

Name	Definition
<code>num_offers_received</code>	$\sum \mathbf{1}\{\text{event} = \text{"offer received"}\}$
<code>num_offers_viewed</code>	$\sum \mathbf{1}\{\text{event} = \text{"offer viewed"}\}$
<code>num_offers_completed</code>	$\sum \mathbf{1}\{\text{event} = \text{"offer completed"}\}$
<code>rate_completed_by_received</code>	<code>num_offers_completed</code> / <code>num_offers_received</code>
<code>rate_viewed</code>	<code>num_offers_viewed</code> / <code>num_offers_received</code>
<code>avg_time_elapsed</code>	Mean(<code>time_completed</code> - <code>time_received</code>) over offers
<code>total_spent</code>	$\sum \text{amount for event} = \text{"transaction"}$
<code>num_transactions</code>	Count of <code>event</code> = "transaction"
<code>avg_spending_per_transaction</code>	<code>total_spent</code> / <code>num_transactions</code>
<code>recency</code>	<code>max(time)</code> - <code>max{time : transaction}</code>
<code>offer_type_{bogo,discount}_received</code>	Count of received offers by type (wide)
<code>offer_type_{bogo,discount}_completed</code>	Count of completed offers by type (wide)
<code>completion_rate_after_info</code>	Completions within 72h of receiving informational offer / <code>#offers</code>

`rate_completed_by_received` and `rate_viewed` with safe handling of zero denominators. Responsiveness over time is captured by computing the elapsed time between `offer_received` and `offer_completed` within each (`customer`, `offer`) pair: we forward-fill `time_received` and backward-fill `time_completed`, then average per customer to obtain `avg_time_elapsed`. Monetary activity is summarized from `transaction` events via `total_spent`, `num_transactions`, and `avg_spending_per_transaction`; recency is defined as the gap between the global maximum event time and each customer’s last transaction time. We then assemble two downstream datasets. The *unsupervised* dataset joins *customers* with behavioral aggregates (`customer_events_summary`, `customer_offer_time_summary`, `customer_spending_summary`), zero-fills non-observed behavior fields, and filters out customers with missing `income`. The *supervised* dataset augments the above with wide matrices of offer-type interactions: counts of *received* and *completed* offers by `offer_type` (e.g., `bogo`, `discount`). Additionally, to proxy the short-term influence of *informational* offers, we compute `completion_rate_after_info` as the fraction of an individual’s completions occurring within 72 hours after receiving an informational offer (with non-applicable cases set to zero). The final supervised table merges all aggregates, fills behavior-derived missing values with zero, retains users with complete core attributes, and is saved for modeling.

4 Unsupervised Part

We apply unsupervised learning to the engineered customer features to discover behavioral segments. The workflow is: (i) prepare a standardized numeric matrix with strict row-wise outlier filtering; (ii) explore structure via correlation analysis and Principal Component Analysis (PCA); (iii) use hierarchical clustering (HC) to bound a plausible range for k ; and (iv) fit k -means in both the standardized space and a reduced PCA space. Cluster assignments are then profiled along visit frequency, basket size, spending, and basic demographics to support interpretation and downstream targeting.

4.1 Data Preparation

To prepare the data for the unsupervised task, from the *unsupervised* table we convert `became_member_on` to `Date`, derive *membership duration* (days since signup), encode `gender` as an ordinal factor (M, F, 0 $\rightarrow$ 1–3), select numeric variables, and standardize them with `scale()`. To stabilize Euclidean distances we implement a strict row-wise IQR filter: values outside $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$ are

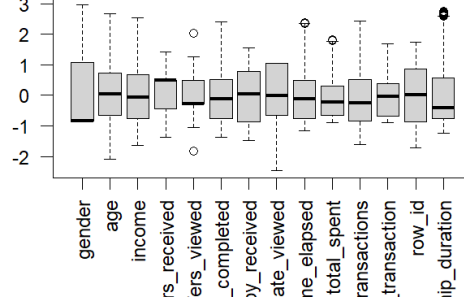


Figure 2: Standardized features after row-wise IQR outlier removal.

flagged per column and any row flagged in *any* column is removed. Boxplots before/after cleaning (Figs. 1–2) show that extreme spikes on spending and transactions are trimmed while central tendency is preserved. The lower-triangle correlation heatmap (Fig. 3) reveals strong associations among engagement/spend variables (`num_offers_completed`, `total_spent`, `num_transactions`) and weak links between demographics and intensity features.

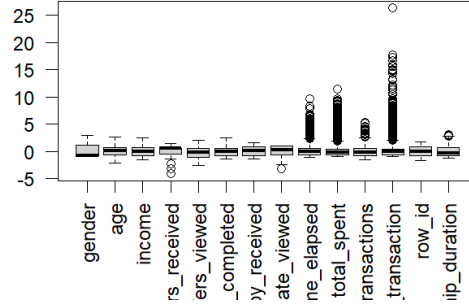


Figure 1: Standardized features before row-wise IQR outlier removal.

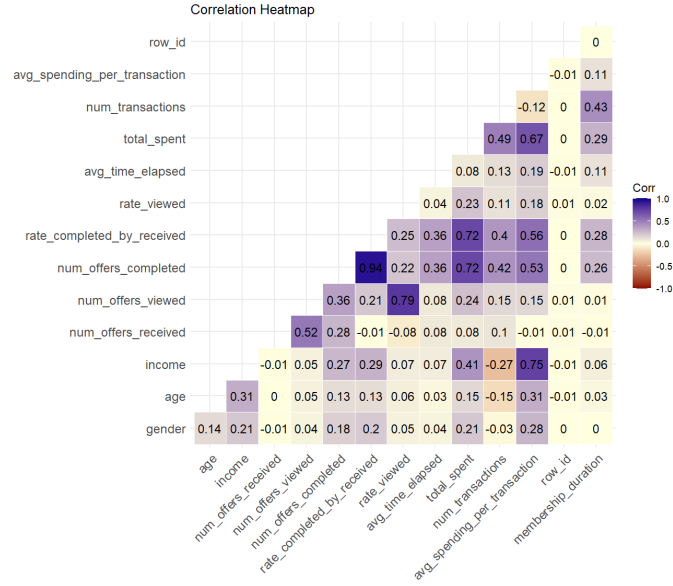


Figure 3: Lower-triangle correlation heatmap of standardized features. Engagement and spend variables form the main correlated block.

4.2 PCA and Hierarchical Clustering (for model selection)

PCA on the cleaned matrix (`prcomp`, no internal scaling) shows a rapid early variance decay with a long tail (Fig. 4); the biplot (Fig. 5) indicates transactional intensity and offer engagement drive the leading directions. Complete-linkage dendrograms on standardized features and on PCA scores (Figs. 6–7) suggest a cut near $h \approx 8.5$, i.e., about six macro-groups.

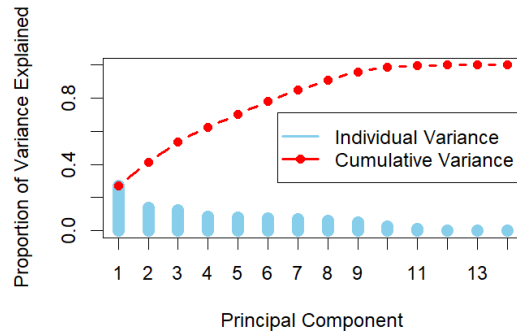


Figure 4: Explained variance by principal component (bars) and cumulative curve (dots/line).

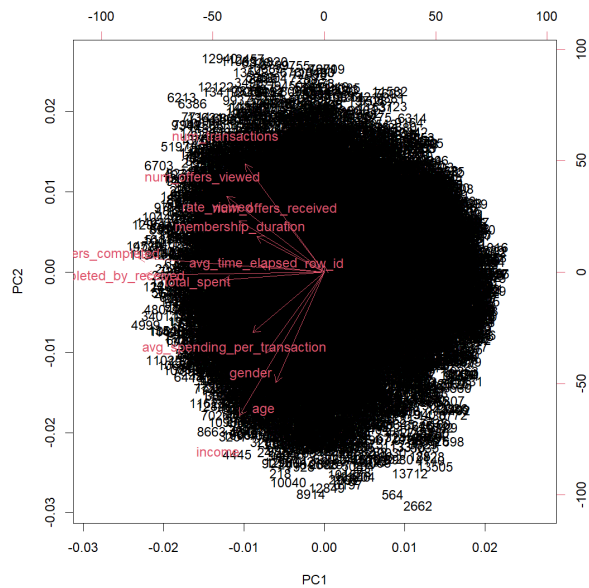


Figure 5: PCA biplot: transactional intensity and offer-engagement variables dominate leading directions.

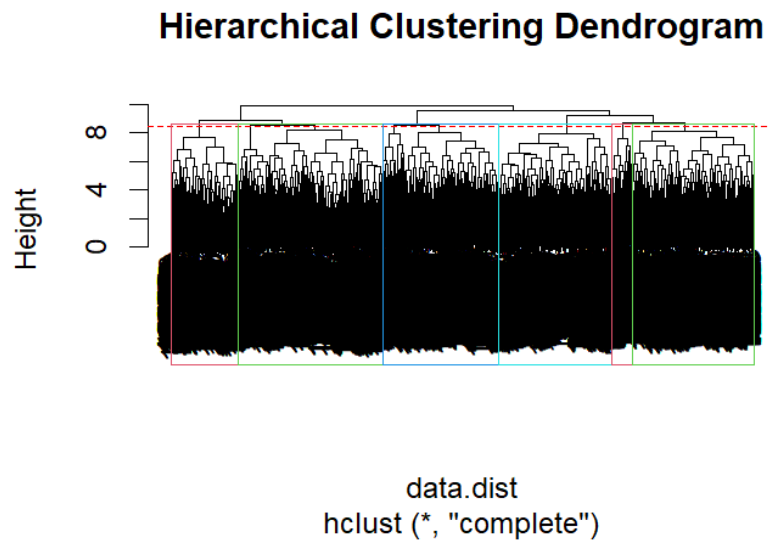


Figure 6: Hierarchical clustering (complete linkage) on standardized features with reference cut at $h \approx 8.5$.

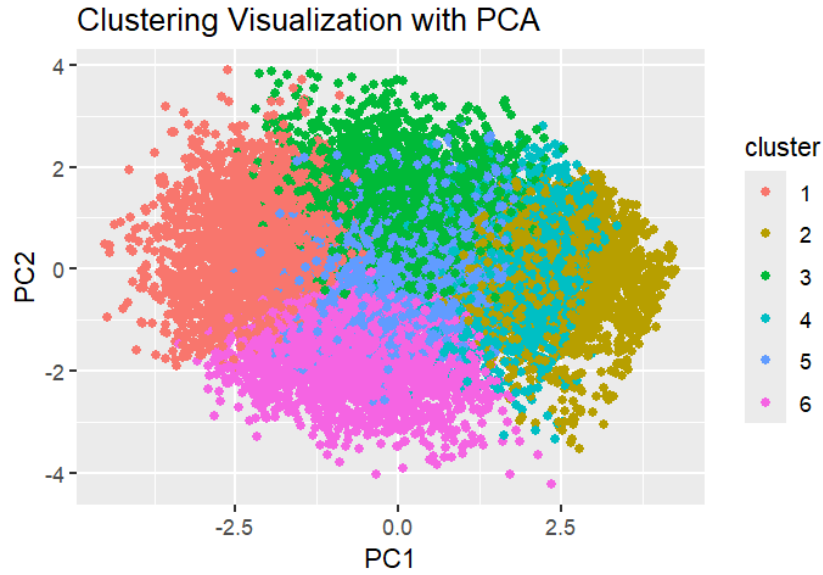


Figure 7: Hierarchical clustering on PCA scores (first components). Similar cut suggests about six macro-groups.

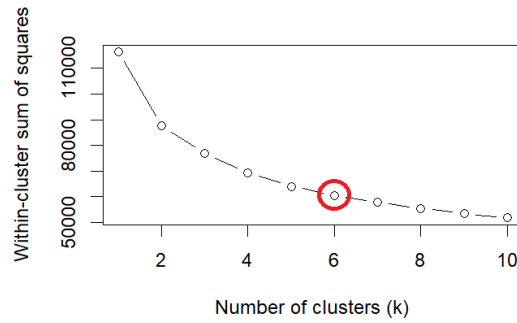


Figure 8: Elbow for k -means on full standardized features; flattening beyond $k \in [5, 6]$.

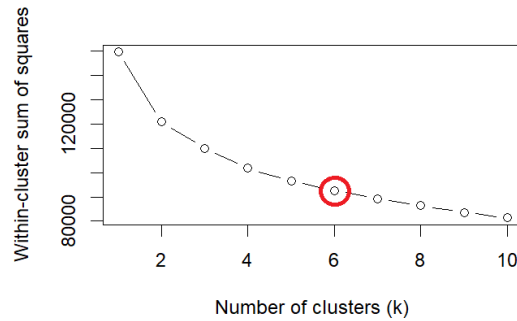


Figure 9: Elbow for k -means on PCA scores (first six PCs); flattening beyond $k \in [5, 6]$.

4.3 K-Means

Since the elbow point is not clearly identifiable, we rely on the hierarchical clustering dendrogram and set $k=6$ for both feature sets. The PC1–PC2 maps (Figs. 10–11) show partially overlapping yet distinguishable regions, consistent with moderate 2D explained variance. Segment profiling reveals clear economic heterogeneity that we show each cluster profile information in the following:

- **Cluster 3 (Customers 19% & Revenue 32%)**
 - Size: 2495 (19% of customers)
 - Revenue: 427171.27 (32% of total)
 - Visits: median 10 (IQR 5); mean 9.78 (sd 3.98)
 - Basket value: median 18.16 (IQR 7.95); mean 18.35 (sd 5.74)
 - Typical spend: median×median 181.57; mean×mean 179.37
 - Mean age: 58
 - Mean income: 71,435
- **Cluster 6 (Customers 15% & Revenue 23%)**
 - Size: 1916 (15% of customers)
 - Revenue: 301840.83 (23% of total)
 - Visits: median 8 (IQR 5); mean 8.45 (sd 3.44)
 - Basket value: median 19.54 (IQR 7.51); mean 19.31 (sd 5.57)
 - Typical spend: median×median 156.33; mean×mean 163.18
 - Mean age: 60
 - Mean income: 75,376
- **Cluster 1 (Customers 15% & Revenue 18%)**
 - Size: 1867 (15% of customers)
 - Revenue: 242465.26 (18% of total)
 - Visits: median 5 (IQR 3); mean 5.76 (sd 2.72)
 - Basket value: median 23.76 (IQR 6.99); mean 23.18 (sd 5.76)
 - Typical spend: median×median 118.81; mean×mean 133.46
 - Mean age: 66
 - Mean income: 90,235
- **Cluster 4 (Customers 18% & Revenue 17%)**
 - Size: 2316 (18% of customers)
 - Revenue: 227890.09 (17% of total)
 - Visits: median 13 (IQR 6); mean 13.44 (sd 4)
 - Basket value: median 7.36 (IQR 8.05); mean 7.77 (sd 4.97)
 - Typical spend: median×median 95.64; mean×mean 104.44
 - Mean age: 42
 - Mean income: 48,180
- **Cluster 2 (Customers 16% & Revenue 6%)**
 - Size: 2101 (16% of customers)

- Revenue: 76297.19 (6% of total)
 - Visits: median 4 (IQR 4); mean 4.3 (sd 3.02)
 - Basket value: median 4.93 (IQR 14.07); mean 9.61 (sd 8.63)
 - Typical spend — median×median 19.71; mean×mean 41.3
 - Mean age: 56
 - Mean income: 62,220
- **Cluster 5 (Customers 17% & Revenue 4%)**
 - Size: 2143 (17% of customers)
 - Revenue: 46678.5 (4% of total)
 - Visits: median 5 (IQR 4); mean 5.38 (sd 3.01)
 - Basket value: median 3 (IQR 1.88); mean 4.11 (sd 4.16)
 - Typical spend: median×median 14.99; mean×mean 22.13
 - Mean age: 47
 - Mean income: 51,031

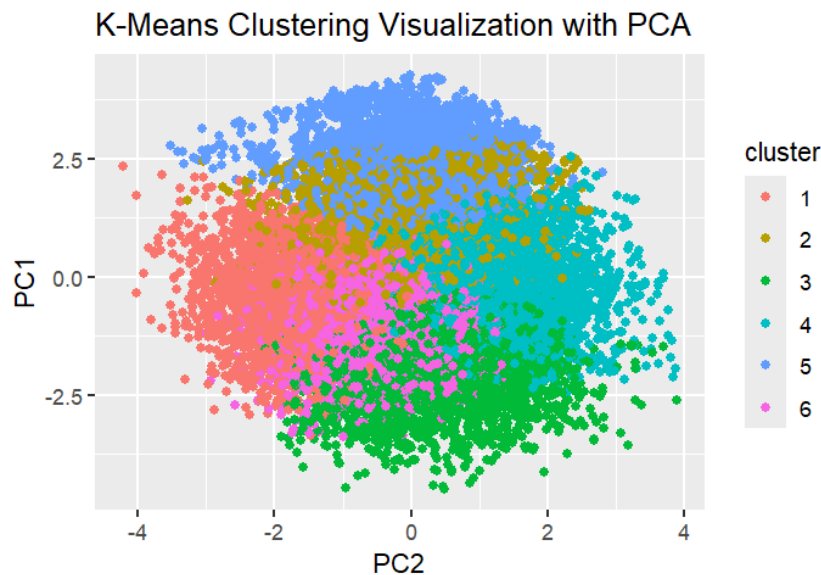


Figure 10: K-means clustering visualization in the PC1–PC2 plane (six clusters).

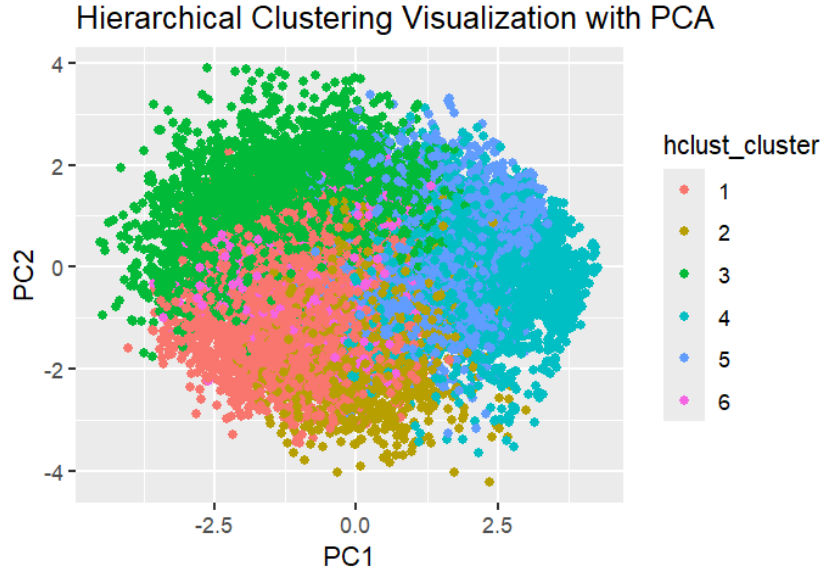


Figure 11: Hierarchical clustering visualization in the PC1–PC2 plane (six clusters).

Customer segmentation reveals six distinct groups differentiated by their spending patterns, visit frequency, and demographics. The key performance measures—visits, basket value, and typical spend—capture how often customers shop, how much they spend per visit, and their overall economic value to the business. Higher “typical spend” (calculated as the product of median visit frequency and basket value) indicates both loyalty and purchasing power.

Cluster 3 stands out as the most valuable segment, generating 32% of total revenue from 19% of customers, with frequent visits and solid basket values—suggesting consistent, high-value shoppers in their late 50s with above-average income. Cluster 6 follows with strong contribution (23% revenue, 15% customers) and similar age-income profiles, representing another premium segment. Cluster 1 includes older, wealthier customers who spend more per basket but visit less often, implying selective yet high-spending behavior. Cluster 4 skews younger (early 40s) and visits frequently but with lower basket values—likely convenience-driven, low-ticket customers. Cluster 2 and Cluster 5 are the least profitable, contributing just 6% and 4% of revenue respectively despite comprising over 30% of the customer base. Their low spend and visit patterns suggest price-sensitive or disengaged shoppers. Overall, the data suggest prioritizing retention and targeted offers for Clusters 3 and 6, while re-engagement strategies might be needed to lift spending from Clusters 2 and 5.

5 Supervised Learning

The supervised learning component of this project aimed to predict each customer’s *preferred offer type*, based on their demographic attributes and behavioral patterns. This target variable represents the offer category (*bogo*, *discount*, or *informational*) that the customer was most responsive to, inferred from historical completion data. The overall workflow involved structured data preprocessing, feature engineering, and a model benchmarking pipeline implemented in R, leveraging the `mlr3` framework to conduct a nested cross-validation for model selection and unbiased performance evaluation.

5.1 Data Preparation

The raw dataset, `supervised.csv`, was loaded and cleaned prior to modeling. The main steps of data preparation are outlined below:

- **Date Conversion:** The variable `became_member_on` was converted to the `Date` type to compute temporal features.
- **Gender Encoding:** The categorical variable `gender` was converted to a factor with levels M, F, and 0. One-hot encoding was then applied via `model.matrix()`, generating dummy variables (`genderM`, `genderF`, `gender0`), after which the original `gender` column was dropped.
- **Membership Duration:** A numerical feature, `membership_duration`, was created to represent the number of days between the member’s joining date and a reference date (15 September 2024). To capture different membership tenures, this feature was binned into three quantile-based categories: *short-term*, *medium-term*, and *long-term*.
- **Standardization:** A z-score standardized version of the membership duration, `membership_duration_zscore`, was computed to normalize scale-sensitive variables.
- **Age and Income Binning:** The numerical variables `age` and `income` were binned into ordered categories:
 - `age_group`: *young adult* (below 25), *middle-aged* (25–45), *senior* (45–65), and *elder* (65+)
 - `income_group`: *low*, *medium*, and *high*, defined by income quantiles

Both resulting categorical variables were ordinal-encoded for compatibility with downstream models. The original numeric `age` and `income` columns were then dropped.

- **Target Variable Definition:** The target variable `preferred_offer_type` was derived by identifying the offer type with the maximum completion count across three columns: `offer_type_bogo_completed`, `offer_type_discount_completed`, and `num_completions_after_info`. Missing values were imputed as zero. The final label had three classes: *bogo*, *discount*, and *informational*.
- **Feature Leakage Prevention:** To avoid target leakage, features directly related to offer completions and counts (`num_offers_received`, `num_offers_viewed`, etc.) were excluded from the modeling dataset.

The resulting dataset, referred to as `supervised_noleak`, was used for model training and evaluation.

5.2 Model Benchmarking and Selection

Model benchmarking was performed through a nested cross-validation procedure using the `mlr3` ecosystem. The data was first split into an 80/20 stratified partition, maintaining class balance across the training and test sets.

The inner loop (3-fold CV) was used for hyperparameter tuning via random search, while the outer loop (5-fold CV) provided an unbiased estimate of generalization performance. The preprocessing pipeline included numerical mean imputation, categorical mode imputation, factor fixing, encoding, constant feature removal, and standard scaling.

Four classifiers were benchmarked:

- **XGBoost (Gradient Boosted Trees)**
- **Random Forest (Ranger Implementation)**
- **Multinomial Logistic Regression**
- **Support Vector Machine (RBF Kernel)**

Each learner was optimized over its respective hyperparameter search space. The XGBoost model achieved the highest mean outer-CV accuracy (**0.9167**), followed by Random Forest (**0.9123**), SVM (**0.8736**), and Multinomial Logistic Regression (**0.8638**). The optimal XGBoost configuration was:

```
xgb.nrounds = 415
xgb.eta = 0.01198
xgb.max_depth = 7
xgb.subsample = 0.777
xgb.colsample_bytree = 0.994
xgb.min_child_weight = 2.33
xgb.gamma = 2.08
```

5.3 Final Model Evaluation

The tuned XGBoost model was retrained on the full 80% training set and evaluated once on the 20% external test set. The performance metrics were as follows:

- **Accuracy:** 0.9157
- **Macro Precision:** 0.9086
- **Macro Recall:** 0.6068
- **Macro F1:** 0.9094

Class-level performance showed strong predictive ability for *bogo* and *discount* offers, while *informational* offers were rarely predicted, indicating a severe class imbalance.

Table 3: Confusion Matrix on Test Set

True Class	Predicted: bogo	Predicted: discount	Predicted: informational
bogo	1756	127	0
discount	121	958	0
informational	1	1	0

Overall, the XGBoost classifier demonstrated robust predictive power and consistency across folds, achieving over 91% accuracy. The poor recall for the *informational* class reflects its underrepresentation in the dataset, which could be addressed through resampling or cost-sensitive adjustments in future iterations.

6 Conclusion and Findings

This project followed an end-to-end data science process, from data preprocessing to the extraction of business insights through unsupervised and supervised learning. The preprocessing phase was fundamental to ensuring the analytical soundness of the results. It involved handling missing and inconsistent data, standardizing variable formats, and constructing derived features to capture behavioral dimensions such as visit frequency and spending intensity. These steps significantly improved data quality and interpretability, forming a solid foundation for subsequent analyses.

Through unsupervised learning, the customer base was segmented into meaningful clusters that revealed distinct behavioral profiles. The analysis showed that while most customers generated modest revenue with relatively few visits, a smaller cluster, accounting for about 19% of customers, contributed over 30% of total revenue. This segment was characterized by high visit frequency and consistent spending, suggesting strong brand engagement and loyalty. Conversely, low-activity

clusters displayed sporadic interactions and lower average spend, indicating either low attachment to the brand or limited perceived value. These insights underline the need for differentiated strategies: retention efforts should focus on the high-value cluster through personalized offers or loyalty rewards, while reactivation campaigns could target low-engagement groups with tailored promotions.

In the supervised phase, predictive models were employed to anticipate customer responses and identify drivers of conversion. The results highlighted the significance of variables such as historical engagement and membership duration in predicting response behavior. This suggests that customers with longer relationships and higher interaction levels are more likely to respond positively to campaigns, providing a data-backed rationale for focusing marketing resources on these segments. The integration of both unsupervised and supervised approaches thus provided a dual perspective, understanding current customer structures while also forecasting future behaviors.

Overall, the project demonstrated how structured data processing and combined analytical methods can translate raw customer data into actionable knowledge. The findings emphasize that strategic prioritization of high-value customers, predictive targeting of responsive groups, and the re-engagement of inactive segments could collectively enhance both revenue efficiency and customer retention. Beyond technical execution, the project reinforced the importance of data-driven reasoning as a foundation for smarter, evidence-based business strategy.

source code:<https://github.com/mohmmadhadi/mavenrewardschallenge->